SITE AUDIT

Website Technical Audit

Precision Structures Inc.

PREPARED FOR	PREPARED BY	DATE	DOCUMENT
Precision Structures Inc.	EBYG Automation — SSC Digital Services	March 19, 2026	01 of 17

Prepared: March 18, 2026 **Auditor:** SSC Digital Services **Subject:** precisionstructures.net — Full technical and content audit **Status:** Pre-launch / Active Development

1. Tech Stack Summary

LAYER	TECHNOLOGY	NOTES
Markup	HTML5	Semantic elements, no templating engine
Styling	CSS3 (single file, 693 lines)	CSS custom properties for theming; responsive breakpoints at 900px and 600px
JavaScript	Vanilla JS (main.js 162 lines, chatbot.js 321 lines)	No frameworks, no bundler, no transpiler
Fonts	Google Fonts — Oswald, Barlow, Source Serif 4	Loaded via external stylesheet link
Hosting	Azure Static Web Apps	GitHub Actions CI/CD pipeline
Source Control	GitHub	Branching: `develop` (staging) / `master` (production)
Build Tools	**None**	Files are edited and deployed directly — no compilation step

Key takeaway: This is the simplest possible production web stack. There is no build step, no dependency tree, and no framework lock-in. This is an advantage for maintainability but a limitation for scalability.

2. Routing / Page Structure Summary

The site consists of **6 static HTML pages** with no client-side routing or server-side rendering.

PAGE	FILE	PURPOSE	KEY COMPONENTS
Home	`index.html`	Landing page / first impression	Hero section, stats strip, about snippet, services cards, process steps, inspirational quote, service area map reference, AI tools promotion
About	`about.html`	Company background	Company history narrative, 5 core values, timeline (1990-2025), key statistics
Services	`services.html`	Service offerings detail	3 service cards (overview), detailed sections for roof trusses / floor trusses / custom structural, SVG technical diagrams, process steps, service area
Gallery	`gallery.html`	Project portfolio	Category filter buttons, 12 gallery items with lightbox overlay, image grid
Resources	`resources.html`	Tools and education	3 AI tool cards, estimate calculator, project planning guide, terminology glossary with SVG diagram, 6 common truss type SVG illustrations
Contact	`contact.html`	Lead capture / inquiry	Contact form (7 fields + file upload), contact info card, embedded Google Maps, 3-column contact method display

Navigation is consistent across all pages: top utility bar (phone, email, social links), main nav with logo, and a mobile hamburger menu at smaller breakpoints.

3. CMS / Admin / Editability Status

There is no CMS, no admin panel, and no content management layer of any kind.

All content is hardcoded directly in HTML files. This has the following consequences:

- **Contact information** (phone number, email addresses, physical address, business hours) is duplicated across **10+ locations** in 6 HTML files. A single phone number change requires editing every file.
- **Service descriptions, stats, company history, and all other copy** live in raw HTML. There is no structured data source, no JSON content file, and no template variable system.
- **Any content update requires a developer** who can edit HTML, commit to Git, and push to trigger deployment.
- **Non-technical staff cannot make updates** without developer assistance.

This is the single largest operational risk for long-term maintainability.

4. Deployment Architecture Summary

```
Developer pushes code
        |
        v
GitHub Repository
        |
        +---> push to `develop` ---> GitHub Actions ---> Azure SWA (staging environment)
        |
        +---> push to `master`  ---> GitHub Actions ---> Azure SWA (production)
```

- **CI/CD:** GitHub Actions workflow with conditional deployment environment based on branch.
- **Hosting:** Azure Static Web Apps (serverless static hosting with global CDN).
- **SSL:** Provided automatically by Azure SWA.
- **Custom domain:** Configured at the Azure SWA level.
- **No backend services** are deployed alongside the static site (no Azure Functions, no database, no server).

The deployment pipeline is clean and functional. Staging and production are properly separated.

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5. Current SEO Status

What exists:

ITEM	STATUS	NOTES
Meta descriptions	Present on all 6 pages	Appear well-written and page-specific
Open Graph tags	Present	Enables social media link previews
`robots.txt`	Present	Standard allow-all configuration
`sitemap.xml`	Present	Lists all 6 pages
Semantic HTML	Partial	Uses some semantic elements but not comprehensively
Responsive design	Yes	Mobile-friendly, passes basic mobile usability

What is missing or broken:

ITEM	IMPACT	PRIORITY
Structured data (JSON-LD)	Google cannot extract business info, services, or reviews for rich snippets	High
Canonical tags	Risk of duplicate content signals if pages are accessible via multiple URLs	Medium
Stale sitemap dates	`lastmod` dates all show 2025-01-01 — signals to crawlers that the site is unmaintained	Medium
No Google Search Console	No visibility into indexing status, crawl errors, or search performance	High
No analytics	Zero data on traffic, user behavior, or conversion	Critical
No local business schema	Missing from Google's local pack and knowledge panel results	High
Page speed optimization	No image compression pipeline, no lazy loading, no critical CSS extraction	Low (site is small)

Bottom line: The site has a baseline SEO foundation but lacks the structured data and tracking infrastructure needed to compete in local search for commercial construction services in the 6-state service area.

6. Current Security Posture

What exists:

HEADER / MEASURE	STATUS
`X-Content-Type-Options: nosniff`	Set globally via `staticwebapp.config.json`
`X-Frame-Options: DENY`	Set globally
`Referrer-Policy: strict-origin-when-cross-origin`	Set globally
`Permissions-Policy`	Set globally (restricts camera, microphone, geolocation)
HTTPS	Enforced by Azure SWA
Chatbot API key	Handled via proxy URL pattern — key is not exposed in client-side code

What is missing:

GAP	RISK	PRIORITY
Content Security Policy (CSP)	No protection against XSS injection. Inline scripts and styles have no nonce/hash allowlisting.	Medium
No CAPTCHA on contact form	Bot spam submissions (once the form actually works)	Medium
No rate limiting	Chatbot proxy and contact form have no abuse throttling	Low (volume is low)
No backend authentication	Not currently needed, but relevant if a client portal is built	Future
No Subresource Integrity (SRI)	External font/script resources could theoretically be tampered with	Low

Overall assessment: Security posture is reasonable for a static brochure site. The global headers are well-configured. CSP is the most notable gap. The bigger risk is operational (the non-functional contact form) rather than adversarial.

7. Current Staging / Production Workflow

ASPECT	CURRENT STATE
Branching model	`develop` = staging, `master` = production
Deployment trigger	Push to either branch triggers GitHub Actions
Staging URL	Azure SWA staging environment (separate URL)
Production URL	precisionstructures.net
Merge process	Manual merge from `develop` to `master`
Automated testing	**None** — no linting, no link checking, no visual regression, no accessibility tests
Preview / review	Manual browser inspection of staging URL

The workflow is functional but has **no automated quality gates**. A broken page or a typo can reach production with no warning.

8. Reusable Components Already Available

Despite the lack of a templating system, the following UI patterns are consistently implemented across all pages and could be extracted into reusable components if a templating layer is introduced:

COMPONENT	DESCRIPTION	APPEARS ON
Top utility bar	Phone, email, social media links	All 6 pages
Main navigation	Logo, nav links, mobile hamburger	All 6 pages
Mobile nav overlay	Slide-in menu for small screens	All 6 pages
Footer	Contact info, nav links, social icons, copyright	All 6 pages
Service cards	Icon + title + description card layout	Home, Services
Process steps	Numbered step indicators with descriptions	Home, Services
Stats strip	Animated counter numbers with labels	Home, About
Section headers	Consistent heading + subtitle + divider pattern	All pages
CTA buttons	Consistent button styling with hover states	All pages
SVG diagrams	Technical truss illustrations	Services, Resources

These components are currently copy-pasted across files. They are not templated, not imported, and not centrally managed.

9. Gaps Relative to Requested Features

This section maps each discussed feature to its current state on the site.

REQUESTED FEATURE	CURRENT STATE	GAP DESCRIPTION
PDF page reviews	Not implemented	No mechanism to generate or deliver PDF snapshots of pages
Client update portal	Not implemented	No admin interface; all edits require Git + code knowledge
AI chatbot	Partially built (chatbot.js, 321 lines)	PrecisionChatbot class exists with Claude API proxy, keyword fallback, typing indicator. Needs persona refinement, conversation logging, and reliability hardening.
Company voice / persona	Not documented	No brand voice guide, no system prompt documentation, no tone-of-voice reference
Email-to-staging workflow	Not implemented	No mechanism to receive email-based update requests and translate them to site changes
SEO technical setup	Partial (meta tags + OG exist)	Missing structured data, canonical tags, analytics, Search Console, updated sitemap
Social media support	Links to Facebook + Instagram exist	No content calendar, no posting workflow, no social-specific content creation
Security + testing	Partial (headers exist)	No CSP, no form backend, no CAPTCHA, no automated testing, no accessibility audit
Analytics / reporting	Not implemented	Zero tracking — no pageviews, no conversion data, no user behavior data
Contact form backend	**NOT FUNCTIONAL**	Form shows a success message on submit but data is not sent anywhere. Leads are being lost.

10. Recommended Implementation Path

Path A: Fastest to Value (Ship Critical Fixes First)

PRIORITY	TASK	EFFORT	IMPACT
1	**Fix contact form** — connect to a backend (Azure Function + SendGrid/email)	1-2 days	Critical: currently losing leads
2	**Add Google Analytics (GA4)**	2 hours	Start collecting data immediately
3	**Update copyright to 2026**	15 minutes	Professionalism
4	**Update sitemap.xml dates**	15 minutes	SEO signal
5	**Add JSON-LD structured data** (LocalBusiness, Service)	4 hours	Local search visibility
6	**Replace gallery placeholders** with real project photos	1-2 days (pending photo availability)	Credibility
7	**Add canonical tags** to all pages	1 hour	SEO hygiene
8	**Refine chatbot persona** and system prompt	1 day	Better user experience for AI feature

Total estimated time: 1-2 weeks. This path gets the site to a professional, functional state.

Path B: Best Long-Term Architecture

PRIORITY	TASK	EFFORT	IMPACT
1	Introduce a lightweight templating layer (e.g., Eleventy, or even simple includes via a build script) to eliminate copy-paste duplication	2-3 days	Maintainability
2	Extract contact info, service data, and stats into a structured JSON content file	1 day	Single source of truth
3	Build a simple admin portal (headless CMS like Decap CMS, or custom lightweight UI)	1-2 weeks	Non-technical staff can update content
4	Implement email-to-staging update workflow	2-3 weeks	Streamlined update process
5	Add automated testing (link checking, accessibility, visual regression) to CI/CD	2-3 days	Quality gates
6	Implement CSP headers and form CAPTCHA	1 day	Security hardening

Total estimated time: 4-6 weeks. This path builds infrastructure for the long haul.

Path C: Best Value (High Impact, Low Effort)

TASK	EFFORT	BUSINESS VALUE
Fix contact form backend	1-2 days	Stop losing leads — immediate ROI
Add GA4 analytics	2 hours	Data-driven decisions
SEO structured data + canonical tags	1 day	Better search ranking
Chatbot persona refinement	1 day	Differentiated user experience
PDF page review script (Playwright)	0.5 days	Client review workflow
Social media content calendar setup	1 day	Consistent online presence

Total estimated time: 1 week. Maximum business impact per hour spent.

11. Brittle, Risky, and Incomplete Items to Fix Before Launch

These items represent real operational or reputational risks and should be addressed before the site is promoted publicly.

#	ITEM	SEVERITY	DESCRIPTION
1	**Contact form does not work**	**CRITICAL**	The form displays a success message on submission, but no data is transmitted to any backend. Every form submission is silently lost. Any visitor who fills out the form believes they have contacted the company and will expect a response that will never come. This must be fixed before any marketing or promotion.
2	**Copyright date says 2025**	Medium	Footer displays "2025" — should read "2026". Signals the site is not actively maintained. Appears on all 6 pages.
3	**Gallery is 92% placeholder images**	High	11 of 12 gallery images are AI-generated. A gallery full of fake images undermines credibility. Either replace with real project photos or reduce the gallery to only real images.
4	**No analytics installed**	High	There is zero data collection. No pageviews, no conversion tracking, no user behavior. Flying blind. Cannot measure ROI of any marketing or SEO effort.
5	**No structured data (JSON-LD)**	Medium	Google cannot extract business name, address, phone, services, or service area for rich results or the local pack. Major missed opportunity for a local service business.
6	**Stale sitemap dates**	Medium	All `lastmod` entries in `sitemap.xml` show `2025-01-01`. Search engines use this signal to determine crawl priority. Stale dates may reduce crawl frequency.
7	**Hardcoded content duplication**	Medium	Contact info is copy-pasted across 10+ locations in 6 files. One missed edit creates inconsistency. A phone number change is a 6-file operation with no safety net.
8	**No Content Security Policy**	Low-Medium	The site has other security headers but no CSP. While the attack surface of a static site is small, CSP is considered a baseline security measure.
9	**No automated testing in CI/CD**	Medium	No broken-link checking, no accessibility testing, no visual regression testing. Errors reach production unchecked.
10	**No Google Search Console verification**	Medium	Cannot monitor indexing status, crawl errors, or search performance. Should be set up alongside analytics.

12. "Do Not Overbuild" Advisory

This site serves a structural steel and truss manufacturing company in the Intermountain West. It is a **lead generation brochure site**, not a SaaS product.

Resist the urge to:

- **Add a JavaScript framework** (React, Vue, etc.) — The current vanilla HTML/CSS/JS stack is perfectly appropriate. The site has 6 pages. A framework adds build complexity, dependency maintenance, and developer onboarding cost with zero user-facing benefit.
- **Build a custom CMS from scratch** — If a content management layer is needed, use an existing solution (Decap CMS, Netlify CMS, or a simple JSON file + admin UI). Do not architect a database-backed content system for a 6-page site.
- **Create an AI chatbot** — The site's [illegible] complex conversational AI platform. Keep the scope to answering common questions about services, pricing

ranges, and process. Do not build a booking system, a quoting engine, or a project management interface into it.

- **Add a database** — Unless a client portal with user accounts is built, there is no need for a database. Form submissions can go to email. Analytics come from GA4. Content lives in files.
- **Introduce microservices** — If backend functions are needed (form handler, chatbot proxy, PDF generation), a single Azure Functions app with a few endpoints is sufficient. Do not create separate services, message queues, or event-driven architectures.
- **Spend weeks on performance optimization** — The site is small, static, and served from a CDN. It is already fast. Image optimization and lazy loading are worth doing, but do not invest in critical CSS extraction, code splitting, or service workers.

The right level of engineering for this project:

- Static HTML/CSS/JS with a thin templating layer (optional)
- 1-3 Azure Functions for backend needs (form handler, chatbot proxy, PDF generation)
- GA4 for analytics
- Structured data for SEO
- A simple, documented content update process

Build the minimum that serves the business. Ship it. Iterate based on data.

End of audit. This document should be reviewed alongside the Feature Gap Analysis for implementation planning.